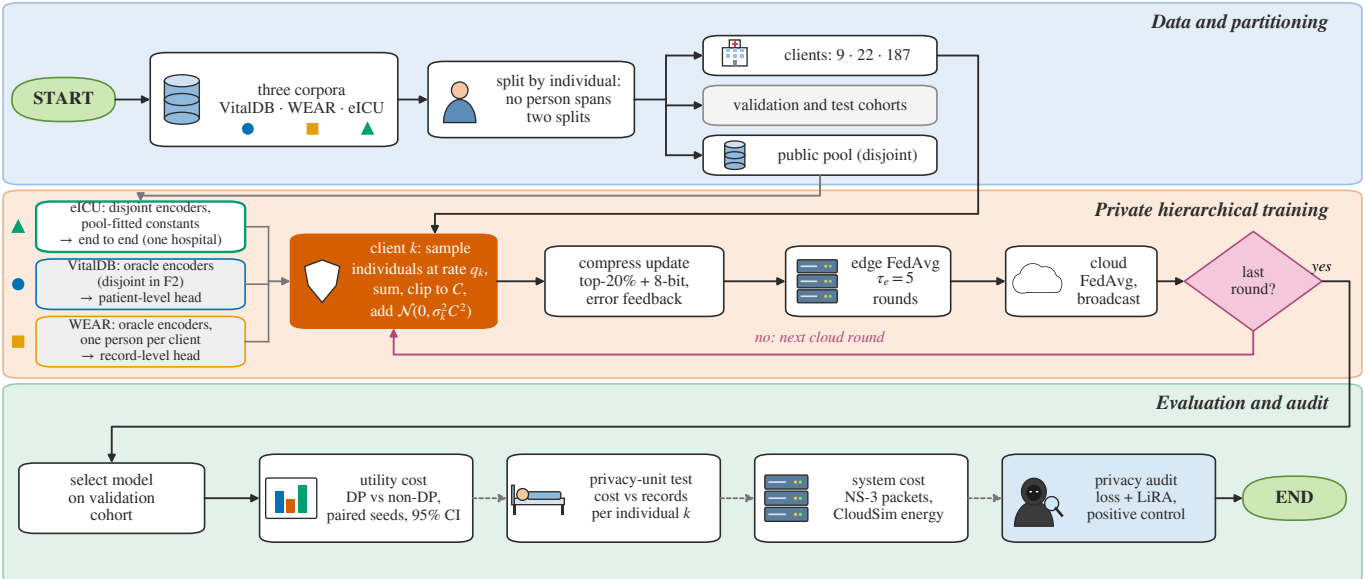
Data and partitioning
START
three corpora
VitalDB · WEAR · eICU
split by individual:
no person spans
two splits
clients: 9 · 22 · 187
validation and test cohorts
public pool (disjoint)
Private hierarchical training
eICU: disjoint encoders,
pool-fitted constants
→ end to end (one hospital)
VitalDB: oracle encoders
(disjoint in F2)
→ patient-level head
WEAR: oracle encoders,
one person per client
→ record-level head
client k: sample
individuals at rate q_k,
sum, clip to C,
add $\mathcal{N}(0, \sigma_k^2 C^2)$
compress update
top-20% + 8-bit,
error feedback
edge FedAvg
$\tau_e = 5$
rounds
cloud
FedAvg,
broadcast
last
round?
yes
no: next cloud round
Evaluation and audit
select model
on validation
cohort
utility cost
DP vs non-DP,
paired seeds, 95% CI
privacy-unit test
cost vs records
per individual k
system cost
NS-3 packets,
CloudSim energy
privacy audit
loss + LiRA,
positive control
END